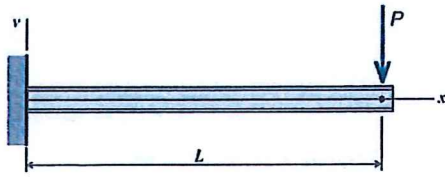


Example: Statically Indeterminate Beam Deflection Problem.

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ENME 202

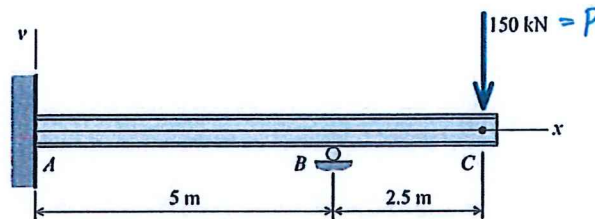
The equation for the vertical deflection, v , and rotation, θ , of a tip-loaded cantilever of length, L , is given by:



$$v = -\frac{Px^2}{6EI}(3L - x)$$

$$\theta = -\frac{Px}{2EI}(2L - x)$$

Q3c: A 7.5m long cantilever beam has a roller support at $x = 5\text{m}$ is subjected to a downwards load of $P = 150\text{kN}$ at the tip.



(2 marks) What is downward deflection of point B from the downwards applied force only, $v_{B,P}$ (ignoring the support at B)? Assume the beam is made of steel ($E = 200\text{GPa}$) and has a second moment of area, $I_z = 640 \times 10^{-6}\text{m}^4$. The formulae above may be useful.

(2 marks) Using the principle of superposition, determine the reaction force at B (the roller support) for the propped cantilever shown.

$$v_{B,P \text{ only}} = \frac{-(150000)(5)^2}{6(200 \times 10^9)(640 \times 10^{-6})}(3(7.5) - 5) = -0.08545\text{m} = -85.45\text{mm}$$

$$v_{B,B_y} = \frac{B_y(5)^2}{6(200 \times 10^9)(640 \times 10^{-6})}(3(5) - 5) = 3.2552 \times 10^{-7} B_y$$

$\Rightarrow$ The total deflection at point B must be zero: $v_{B,P} + v_{B,B_y} = 0$.

$$v_{B,B_y} = 3.2552 \times 10^{-7} B_y + -0.08545 = 0$$

$$\Rightarrow B_y = 262.5\text{kN}.$$

$v_{B,P} = -85.5\text{mm}.$

$B_y = 263\text{kN}.$

Q3d: (3 marks) What is the total deflection of the tip of the propped cantilever beam (point C), based on your solved reaction force, B_y ? (If you were unable to determine the value of B_y , use $B_y = 200\text{kN}$)

$$v_{C,P} = \frac{-(150000)(7.5)^2}{6(200 \times 10^9)(640 \times 10^{-6})}(3(7.5) - 7.5) = -0.1648\text{m}$$

$$v_{B,B_y} = \frac{+(262500)(5)^2}{6(200 \times 10^9)(640 \times 10^{-6})}(3(5) - 5) = 0.08545\text{m} \text{ (or just use result of Q3c above)}$$

$$\theta_{B,B_y} = \frac{+(262500)(5)}{2(200 \times 10^9)(640 \times 10^{-6})}(2(5) - 5) = 0.02563 \text{ radians.}$$

$$v_{C,B_y} = v_{B,B_y} + \theta_{B,B_y} L_{BC}$$

$$= 0.08545 + (0.02563)(2.5) = 0.1495\text{m}$$

$$v_{C,\text{total}} = v_{C,P} + v_{C,B_y} = (-0.1648) + (0.1495) = -0.01528\text{m}$$

$v_{C,\text{total}} = -15.3\text{mm}$

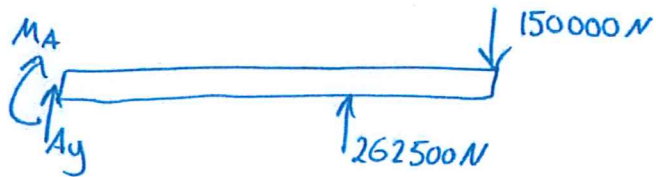
TURN OVER

Q3e: (3 marks) Based on your solved value for B_y , what are the moment and force reaction at A? (If you were unable to determine the value of B_y , use $B_y = 200 \text{ kN}$)

Overall Equilibrium:

$$\uparrow \sum F_y = 0: A_y + 262.5 - 150 = 0$$

$$A_y = -112.5 \text{ kN}$$



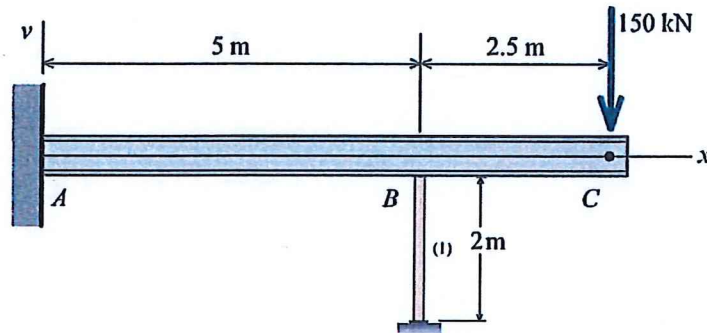
$$\sum M_A = 0: -M_A + (262.5)(5) - (150)(7.5) = 0$$

$$M_A = 187.5 \text{ kNm}$$

$$A_y = -113 \text{ kN.}$$

$$M_A = 188 \text{ kNm}$$

The support condition at point B has been changed to a flexible axial member, as shown below. The axial member is 2m long, has elastic modulus of 70 GPa and a cross-sectional area of 400 mm². The beam properties are unchanged ($E = 200 \text{ GPa}$ and $I_z = 640 \times 10^{-6} \text{ m}^4$).



Q3f: (4 marks) What is the new support reaction, B_y ?

$$v_{B,P} = -0.08545 \text{ m from Q3c}$$

$$v_{B,B_y} = (3.2552 \times 10^{-7}) B_y \text{ from Q3c}$$

$$\text{and } B_y = \frac{-v_{B,\text{total}} A_1 E_1}{L_1} \Rightarrow v_{B,\text{total}} = \frac{-B_y L_1}{A_1 E_1}$$

Based on the flexibility of member (1).
(axial force-deformation relationship)

$$v_{B,P} + v_{B,B_y} = v_{B,\text{total}} = \frac{-B_y L_1}{A_1 E_1}$$

$$-0.08545 + (3.2552 \times 10^{-7}) B_y = \frac{-B_y (2)}{(400 \times 10^{-6})(70 \times 10^9)}$$

$$-0.08545 = (-7.1429 \times 10^{-8} - 3.255 \times 10^{-7}) B_y$$

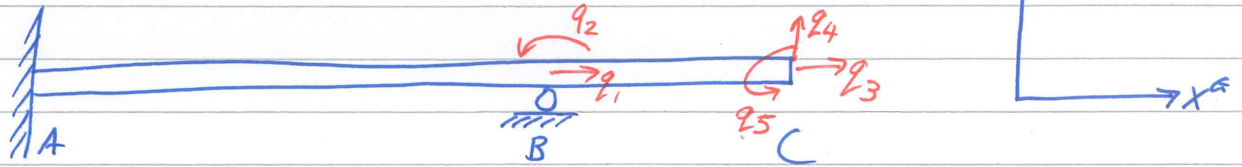
$$\Rightarrow B_y = \frac{-0.08545}{-3.9693 \times 10^{-7}}$$

$$B_y = 215.3 \text{ kN}$$

$$B_y = 215 \text{ kN.}$$

TURN OVER

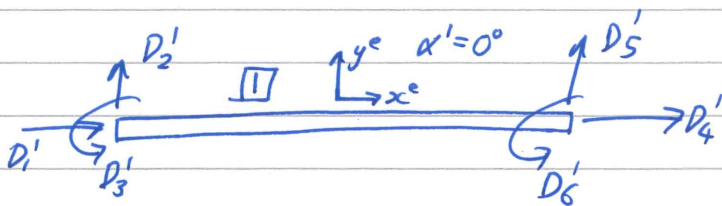
Solve old ENME202 Final Exam Question:



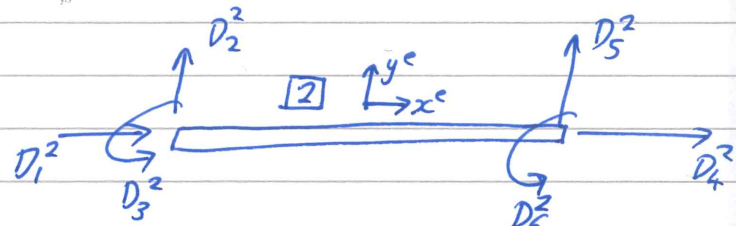
$$I_1 = I_2 = 640 \times 10^{-6} \text{ m}^4.$$

$$E_1 = E_2 = 200 \text{ GPa} = 200 \times 10^9 \text{ Pa}.$$

$$L_1 = 5 \text{ m}, \quad L_2 = 2.5 \text{ m}$$



$$A^1 = \begin{matrix} & \begin{matrix} D_1' & D_2' & D_3' & D_4' & D_5' & D_6' \end{matrix} \\ \begin{matrix} q_1' \\ q_2' \\ q_3' \\ q_4' \\ q_5' \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$



$$A^2 = \begin{matrix} & \begin{matrix} D_1'' & D_2'' & D_3'' & D_4'' & D_5'' & D_6'' \end{matrix} \\ \begin{matrix} q_1'' \\ q_2'' \\ q_3'' \\ q_4'' \\ q_5'' \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$Q = \begin{Bmatrix} 0 \text{ N} \\ 0 \text{ Nm} \\ 0 \text{ N} \\ -150,000 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$$

The original problem did not provide a cross-sectional area for the elements. Let's assume that $A^1 = A^2 = 2 \times 10^{-4} \text{ m}^2$.

L1 = 5 L2 = 2.5

E = 200e9

A = 2e-4

I = 640e-6

K1hat, Lambda1 = Global_FRAME(E, I, A, L1, 0)

K2hat, Lambda2 = Global_FRAME(E, I, A, L2, 0)

K1hat =

1.0e+08 ×

0.0800	0	0	-0.0800	0	0
0	0.1229	0.3072	0	-0.1229	0.3072
0	0.3072	1.0240	0	-0.3072	0.5120
-0.0800	0	0	0.0800	0	0
0	-0.1229	-0.3072	0	0.1229	-0.3072
0	0.3072	0.5120	0	-0.3072	1.0240

Lambda1 =

1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1

K2hat =

1.0e+08 ×

0.1600	0	0	-0.1600	0	0
0	0.9830	1.2288	0	-0.9830	1.2288
0	1.2288	2.0480	0	-1.2288	1.0240
-0.1600	0	0	0.1600	0	0
0	-0.9830	-1.2288	0	0.9830	-1.2288
0	1.2288	1.0240	0	-1.2288	2.0480

Lambda2 =

1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1

A1 = 0 0 0 1 0 0

0 0 0 0 0 1

0 0 0 0 0 0

0 0 0 0 0 0

0 0 0 0 0 0

A2 = 1 0 0 0 0 0

0 0 1 0 0 0

0 0 0 1 0 0

0 0 0 0 1 0

0 0 0 0 0 1

KG1 = A1 @ K1hat @ A1.T

KG2 = A2 @ K2hat @ A2.T

KG1 = 1.0e+08 ×

0.0800	0	0	0	0
0	1.0240	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

KG2 = 1.0e+08 ×

0.1600	0	-0.1600	0	0
0	2.0480	0	-1.2288	1.0240
-0.1600	0	0.1600	0	0
0	-1.2288	0	0.9830	-1.2288
0	1.0240	0	-1.2288	2.0480

KG = KG1 + KG2

KG = 1.0e+08 ×

0.2400	0	-0.1600	0	0
0	3.0720	0	-1.2288	1.0240
-0.1600	0	0.1600	0	0
0	-1.2288	0	0.9830	-1.2288
0	1.0240	0	-1.2288	2.0480

Q =

0 N
0 Nm
0 N
-150000 N
0 Nm

q =

0 m	- the horizontal deflection at B
-0.0037 rad	- the rotation at B
0 m	- the horizontal deflection at C
-0.0153 m	- the vertical deflection at C (tip deflection)
-0.0073 rad	- the rotation at C (tip rotation)

F1 = K1hat @ A1.T @ q

F2 = K2hat @ A2.T @ q

RB = F1[4] + F2[1]

F1 = 1.0e+05 ×

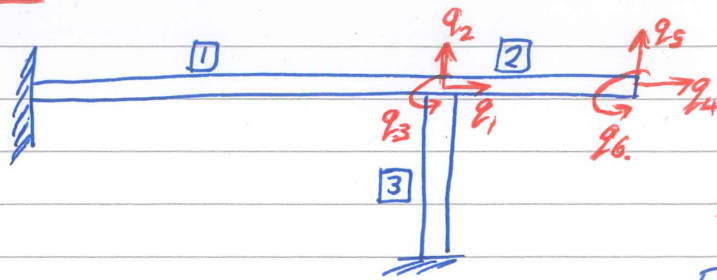
0 N
-1.1250 N - the vertical reaction force at A (A_y)
-1.8750 Nm - the moment reaction at A (M_A)
0 N
1.1250 N
-3.7500 Nm

F2 = 1.0e+05 ×

0 N
1.5000 N
3.7500 Nm
0 N
-1.5000 N
0 Nm

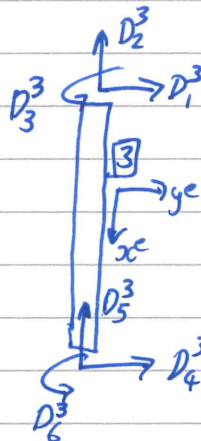
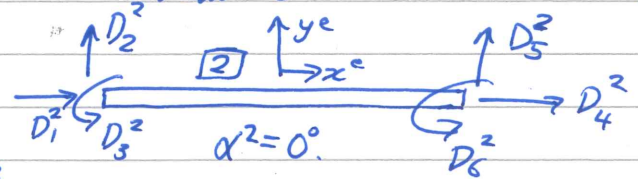
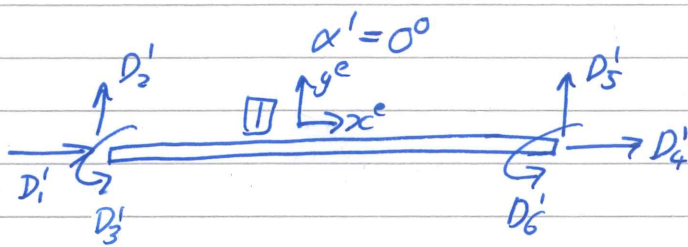
RB = 2.6250e+05N = 262.5kN - the total vertical reaction force at B (B_y)

Alternatively, we can re-work this example with three elements to answer Q3f:



NOTE: We weren't given a I value for element 3. Entering a zero value for I^3 will cause a division by zero issue. Therefore, let's assume a small value of $I^3 = 1 \times 10^{-10} \text{ m}^4$.

Element Freebody Diagrams:



$$\alpha^3 = -90^\circ \text{ or } +270^\circ$$

$$A^1 = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{bmatrix} \begin{bmatrix} D_1^1 & D_2^1 & D_3^1 & D_4^1 & D_5^1 & D_6^1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{bmatrix} \begin{bmatrix} D_1^2 & D_2^2 & D_3^2 & D_4^2 & D_5^2 & D_6^2 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \mathbf{I}_{6 \times 6}$$

$$A^3 = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{bmatrix} \begin{bmatrix} D_1^3 & D_2^3 & D_3^3 & D_4^3 & D_5^3 & D_6^3 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

L1 = 5	L2 = 2.5	L3 = 2m
E1 = 200e9	E2 = 200e9	E3 = 70e9
A1 = 2e-4	A2 = 2e-4	A3 = 400e-6
I1 = 640e-6	I2 = 640e-6	I3 = 1e-10 - assume a small value for I3
K1hat,Lambda1 = Global_FRAME(E1, I1, A1, L1, 0)		
K2hat,Lambda2 = Global_FRAME(E2, I2, A2, L2, 0)		
K3hat,Lambda3 = Global_FRAME(E3, I3, A3, L3, -90)		
K1hat =		
1.0e+08 ×		
0.0800	0	0
0	0.1229	0.3072
0	0.3072	1.0240
-0.0800	0	0
0	-0.1229	-0.3072
0	0.3072	0.5120
Lambda1 =		
1	0	0
0	1	0
0	0	1
0	0	0
0	0	0
0	0	0
K2hat =		
1.0e+08 ×		
0.1600	0	0
0	0.9830	1.2288
0	1.2288	2.0480
-0.1600	0	0
0	-0.9830	-1.2288
0	1.2288	1.0240
Lambda2 =		
1	0	0
0	1	0
0	0	1
0	0	0
0	0	0
0	0	0
K3hat =		
1.0e+07 ×		
0.0000	-0.0000	0.0000
-0.0000	1.4000	0.0000
0.0000	0.0000	0.0000
-0.0000	0.0000	-0.0000
0.0000	-1.4000	-0.0000
0.0000	0.0000	0.0000
Lambda3 =		
0.0000	-1.0000	0
1.0000	0.0000	0
0	0	1.0000
0	0	0
0	0	0
0	0	0
A1 =		
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
A2 =		
1	0	0
0	1	0
0	0	1
0	0	0
0	0	0
0	0	0

A3 =

1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

KG1 = A1 @ K1hat @ A1.T

KG2 = A2 @ K2hat @ A2.T

KG3 = A3 @ K3hat @ A3.T

KG1 = 1.0e+08 ×

0.0800	0	0	0	0	0
0	0.1229	-0.3072	0	0	0
0	-0.3072	1.0240	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

KG2 = 1.0e+08 ×

0.1600	0	0	-0.1600	0	0
0	0.9830	1.2288	0	-0.9830	1.2288
0	1.2288	2.0480	0	-1.2288	1.0240
-0.1600	0	0	0.1600	0	0
0	-0.9830	-1.2288	0	0.9830	-1.2288
0	1.2288	1.0240	0	-1.2288	2.0480

KG3 = 1.0e+07 ×

0.0000	0.0000	0.0000	0	0	0
0.0000	1.4000	-0.0000	0	0	0
0.0000	-0.0000	0.0000	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

KG = KG1 + KG2 + KG3

KG = 1.0e+08 ×

0.2400	0.0000	0.0000	-0.1600	0	0
0.0000	1.2459	0.9216	0	-0.9830	1.2288
0.0000	0.9216	3.0720	0	-1.2288	1.0240
-0.1600	0	0	0.1600	0	0
0	-0.9830	-1.2288	0	0.9830	-1.2288
0	1.2288	1.0240	0	-1.2288	2.0480

Q =

0
0
0
0
-150000
0

q =

0.0000 m	- the horizontal deflection at B
-0.0154 m	- the vertical deflection at B
-0.0083 rad	- the rotation at B
0.0000 m	- the horizontal deflection at C
-0.0422 m	- the vertical deflection at C (transverse tip deflection)
-0.0119 rad	- the rotation at C (tip rotation)

Calculate the element reaction forces in global co-ordinates		
$F1 = K1hat @ A1.T @ q$ $F1 = \begin{matrix} 1.0e+05 \times \\ 0 \\ -0.6526 \\ 0.4868 \\ 0 \\ 0.6526 \\ -3.7500 \end{matrix}$	$F2 = K2hat @ A2.T @ q$ $F2 = \begin{matrix} 1.0e+05 \times \\ 0 \\ 1.5000 \\ 3.7500 \\ 0 \\ -1.5000 \\ -0.0000 \end{matrix}$	$F3 = K3hat @ A3.T @ q$ $F3 = \begin{matrix} 1.0e+05 \times \\ -0.0000 \\ \underline{-2.1526} \\ -0.0000 \\ 0.0000 \\ \underline{2.1526} \\ -0.0000 \end{matrix}$ The axial force within member 3 is 215.26kN (compression)